

A Sharp M-Matrix Threshold for a Two-Dimensional Oblique Skorokhod Map

Route-A source-local certificate HCS-C346

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Abstract

For oblique regulation in the quadrant with the stated two-axis reflection matrix R , we prove the *fixed point and sharp M-matrix chamber*: every càdlàg input has one and only one regulated path precisely when $\rho\sigma < 1$.

1 Convention and the sharp existence theorem

Fix $T < \infty$, $\rho, \sigma \geq 0$, and

$$R = \begin{pmatrix} 1 & -\rho \\ -\sigma & 1 \end{pmatrix}.$$

Let $x = (x_1, x_2) \in D([0, T], \mathbb{R}^2)$ with $x(0) \geq 0$ coordinatewise. A solution consists of càdlàg z, y such that

$$z = x + Ry \geq 0, \quad y(0) = 0, \quad y_i \text{ is nondecreasing}, \quad \int_{[0, t]} z_i(s) dy_i(s) = 0. \quad (1)$$

The integrand is the *post-jump* state. Thus a regulator jump is supported on an axis after the jump, including simultaneous corner jumps.

For scalar càdlàg f , write

$$(\mathcal{L}f)(t) = \sup_{0 \leq s \leq t} [-f(s)]_+.$$

Lemma 1 (One-sided regulator). *If $f(0) \geq 0$, the unique nondecreasing càdlàg r , with $r(0) = 0$, for which $g = f + r \geq 0$ and $\int g dr = 0$ is $r = \mathcal{L}f$. Moreover,*

$$\|\mathcal{L}f - \mathcal{L}h\|_\infty \leq \|f - h\|_\infty. \quad (2)$$

Proof. The running supremum is the least nondecreasing correction making $f + r$ nonnegative. It increases only at a new negative running minimum, where the post-jump corrected state is zero. Conversely, feasibility forces every regulator to dominate $\mathcal{L}f$. A first strict excess would be an increase with positive corrected state, contradicting complementarity. Finally, positive part and running supremum are one-Lipschitz, proving (2). $\square$

Theorem 2 (Sharp all-input threshold). *Problem (1) has exactly one solution for every admissible input and every finite horizon if and only if*

$$\rho\sigma < 1. \quad (3)$$

Inside this chamber, its regulator is the unique solution of

$$y_1 = \mathcal{L}(x_1 - \rho y_2), \quad y_2 = \mathcal{L}(x_2 - \sigma y_1). \quad (4)$$

Proof. With the other coordinate held fixed, each line of (1) is the scalar problem in Lemma 1; hence (1) and (4) are equivalent. Conversely, at time zero the fixed-point equations and $x(0) \geq 0$ imply $y_1(0) \leq \rho y_2(0)$ and $y_2(0) \leq \sigma y_1(0)$; since $\rho\sigma < 1$, both values vanish. Thus the fixed point has the required initial condition.

First suppose $\rho, \sigma > 0$, put $w_1 = \sqrt{\rho}$, $w_2 = \sqrt{\sigma}$, $q = \sqrt{\rho\sigma}$, and define

$$\|u\|_w = \max_{i=1,2} \frac{\|u_i\|_\infty}{w_i}.$$

For the right side Φ_x of (4), inequality (2) gives

$$\|\Phi_x(u) - \Phi_x(v)\|_w \leq q\|u - v\|_w.$$

Thus (3) makes Φ_x a contraction on the complete space of bounded càdlàg pairs. Its unique fixed point is càdlàg and solves the problem. The constant q is the best uniform coefficient: each weighted cross-coordinate estimate can be attained by a constant difference after a suitable downward input shift.

If $\rho = 0$, solve successively $y_1 = \mathcal{L}x_1$ and $y_2 = \mathcal{L}(x_2 - \sigma y_1)$; if $\sigma = 0$, reverse the order. This also covers $\rho = \sigma = 0$ and proves sufficiency.

For necessity, let $\rho\sigma = 1$. Then $R(\rho, 1)^\top = 0$, so zero input has the distinct solutions

$$y = (\rho h, h), \quad z = 0, \tag{5}$$

for every nondecreasing càdlàg h with $h(0) = 0$. More strongly, whenever $\rho\sigma \geq 1$, an input jumping from zero to $(-1, -1)$ cannot be feasible: the post-jump values would obey

$$y_1 \geq 1 + \rho y_2, \quad y_2 \geq 1 + \sigma y_1,$$

and hence $(1 - \rho\sigma)y_1 \geq 1 + \rho$, an impossibility. Therefore no point on or beyond the wall is well posed for every input. $\square$

Source lineage

This source-local reconstruction makes no literature-priority claim.

References

- [1] J. M. Harrison and M. I. Reiman, “Reflected Brownian Motion on an Orthant,” *Annals of Probability* 9 (1981), 302–308. DOI: [10.1214/aop/1176994471](https://doi.org/10.1214/aop/1176994471).
- [2] P. Dupuis and H. Ishii, “On Lipschitz Continuity of the Solution Mapping to the Skorokhod Problem, with Applications,” *Stochastics and Stochastics Reports* 35 (1991), 31–62. DOI: [10.1080/17442509108833688](https://doi.org/10.1080/17442509108833688).